

The Alignment Trap: Structural Limits of SNOVA-Style Matrix Signatures

Justin Thaler* Kai Zhang[†] Scott Kominers[‡] Quang Dao[§]

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Abstract

Matrix-based MQ signatures can compress public keys by choosing secret block coefficients that align with public left and right multiplier algebras. We show that full-capacity alignment is rigid. If the multiplier algebras are degree- d fields and a d -dimensional linear secret support commutes with them on the two sides, then the support and the multiplier fields are forced to be one adjoint pair. Allowing secret coefficients to relabel multipliers by field automorphisms does not escape: every nonzero linear support contained in a field normalizer lies in one Frobenius-semilinear coset. Even linearity is not essential: a sufficiently large arbitrary two-sided normalizer support forces the same adjoint pair. Replacing the field by a genuinely noncommutative irreducible algebra only shrinks the dimension of its commuting multiplier algebra.

For odd characteristic, the adjoint pair induces a canonical involution on the d^2 ordered left/right labels. Symmetric base forms factor through a Hermitian quotient of dimension $d(d+1)/2$, while skew-symmetric forms factor through the complementary anti-Hermitian quotient of dimension $d(d-1)/2$. A transpose-stable w -dimensional base-form family therefore cannot realize wd^2 independent ordered features. Achieving the raw count requires a maximally non-self-adjoint family whose symmetric and skew projections are both injective and whose two polynomial image channels are full and disjoint.

These results give a precise trilemma for SNOVA-style designs: accept the Hermitian capacity loss, restore a full anti-Hermitian/exterior channel, or abandon the alignment-based congruence-reuse mechanism. The theorem does not claim that every construction in the latter two branches is insecure. It shows that none is a local repair preserving the present alignment philosophy and its ordered-label security count. As an application, the symmetric $q = 19$ SNOVA Version 2.3 forms collapse 16 labels to 10 when $d = 4$ and 4 labels to 3 when $d = 2$; the resulting concrete all-parameter forgery is developed separately in the companion cryptanalysis.

1 Introduction

SNOVA-style matrix signatures obtain compact public descriptions by reusing a small family of congruence-transformed quadratic forms. Their public maps are expanded by multiplying matrix-valued variables on the left and right by members of a small algebra. A security analysis may then count the ordered left/right labels as distinct quadratic directions. The same algebraic alignment that makes the public-key derivation work, however, constrains which label spaces can genuinely remain independent.

*a16z crypto research and Georgetown University

[†]MIT

[‡]Harvard University and a16z crypto research

[§]Carnegie Mellon University and LayerZero Labs

This paper isolates that architectural constraint. It deliberately does not repeat the concrete forgery analysis of SNOVA Version 2.3; that analysis is given in the companion paper [1]. Here the question is broader and structural:

How much left/right feature capacity can a matrix-based MQ design retain while its secret coefficients align with the public multiplier algebras strongly enough to reuse one congruence-transformed family of forms?

Our first answer is a capacity theorem. Let $\mathcal{T}$ be an irreducible secret algebra acting on $\mathbb{F}_q^d$, and let D be its centralizer. The finite double-centralizer theorem forces

$$D \cong \mathbb{F}_{q^e}, \quad \mathcal{T} \cong \text{Mat}_{d/e}(\mathbb{F}_{q^e})$$

for a divisor $e \mid d$. Every pointwise aligned multiplier algebra lies in D or $D^\top$, so it has dimension at most e . The field case $e = d$ is therefore the unique extremum retaining d multiplier dimensions on each side. Moving to a genuinely noncommutative secret algebra enlarges the secret algebra only by shrinking the public multiplier capacity.

At full multiplier dimension the rigidity is stronger. Degree- d field algebras in $\text{Mat}_d(\mathbb{F}_q)$ are self-centralizing. Hence a full d -dimensional pointwise aligned secret support is exactly the right multiplier field and the transpose of the left multiplier field. This does not assume that the support is an algebra, is irreducible, or is closed under multiplication. Automorphism relabeling also fails to create a new degree of freedom: a linear support whose nonzero elements normalize a degree- d field is contained in one Frobenius-semilinear coset. We further give a cardinality criterion showing that sufficiently large nonlinear supports are equally rigid.

Our second answer concerns quadratic features. Once alignment forces the left and right fields to be an adjoint pair, scalar transposition defines a canonical involution on their ordered tensor labels. Symmetric base forms see only the Hermitian quotient; skew-symmetric forms see only the anti-Hermitian quotient. This yields the exact capacities $d(d+1)/2$ and $d(d-1)/2$. For a family W of base matrices, the total rank is controlled by the ranks of the symmetric and skew projections of W . A transpose-stable family always loses capacity. Recovering all ordered labels requires $W \cap W^\top = 0$, full symmetric and skew projections, and disjoint polynomial image channels. Thus “make the forms slightly nonsymmetric” is not a local repair.

The resulting conclusion is a trilemma rather than a universal impossibility claim. A design must accept the Hermitian cap, deliberately restore the full anti-Hermitian channel, or change the alignment and key-derivation mechanism. The second and third branches may contain secure constructions, but they are architecturally different from the symmetric aligned design whose security is based on counting ordered labels.

Contributions.

1. We classify the multiplier capacity of irreducible aligned secret algebras by the finite double-centralizer theorem.
2. We prove rigidity for pointwise alignment, semilinear normalizer alignment, and sufficiently large nonlinear supports.
3. We derive the Hermitian and anti-Hermitian quotient dimensions for an arbitrary adjoint pair of multiplier fields.
4. We quantify the nonsymmetry necessary to recover the raw ordered-label count and formulate the resulting no-local-repair trilemma.
5. We specialize the theorem to the symmetric public forms of SNOVA Version 2.3 and connect the structural loss to the companion concrete cryptanalysis.

Scope. The alignment theorems hold over arbitrary finite fields; the decomposition into symmetric and anti-Hermitian channels uses odd characteristic. We do not assert that every maximally non-self-adjoint or non-aligned descendant is broken. We prove that such a descendant has left the design regime in which full alignment, transpose-stable forms, and a d^2 ordered-label count coexist.

2 Model and notation

Let $V = \mathbb{F}_q^d$ and identify $\text{End}_{\mathbb{F}_q}(V)$ with $\text{Mat}_d(\mathbb{F}_q)$. A *degree- d field algebra* is an embedded copy $A \cong \mathbb{F}_{q^d}$ in $\text{Mat}_d(\mathbb{F}_q)$. For a subalgebra B , write

$$\mathcal{C}(B) = \{X : XB = BX\}$$

for its centralizer and $N(B)$ for the normalizer of $B^\times$ in $\text{GL}_d(\mathbb{F}_q)$.

A right multiplier algebra A_R and a left multiplier algebra A_L act on a base matrix P through ordered labels $\xi \otimes \eta \in A_L \otimes A_R$:

$$\Phi_P(\xi \otimes \eta)(u) = u^\top \Sigma_\xi P \Sigma_\eta u.$$

The block embeddings Σ_ξ do not affect the local $d \times d$ algebraic arguments, so we suppress their ambient identity factors when studying alignment.

Pointwise alignment means that each secret block coefficient t satisfies

$$t^\top a_L = a_L t^\top, \quad a_R t = t a_R$$

for all public multipliers. Semilinear alignment relaxes commutation to normalization, allowing conjugation by t to relabel field elements by a Frobenius automorphism. These are exactly the identities needed to move public multipliers past secret coefficients while keeping one congruence-transformed base-form family reusable.

3 Alignment capacity and rigidity

The public-key derivation uses two properties of the alignment algebra at once. Secret block coefficients commute with the public right multipliers, and their transposes commute with the public left multipliers. These are not basis conventions: with full-dimensional support, they determine the two multiplier algebras.

The first theorem removes a possible escape before specializing to a field. It is a finite double-centralizer statement: a noncommutative irreducible secret algebra can be large only by making its commuting multiplier algebra small.

Theorem 3.1 (Alignment capacity and the field extremum). *Let $\mathcal{T} \subseteq \text{Mat}_d(\mathbb{F}_q)$ be a unital algebra acting irreducibly on $V = \mathbb{F}_q^d$, and put $D = \mathcal{C}(\mathcal{T})$. Then there is a divisor $e \mid d$ such that*

$$D \cong \mathbb{F}_{q^e}, \quad \mathcal{T} = \mathcal{C}(D) \cong \text{Mat}_{d/e}(\mathbb{F}_{q^e}), \quad \dim_{\mathbb{F}_q} \mathcal{T} = \frac{d^2}{e}. \quad (1)$$

If public right and left multiplier algebras satisfy

$$A_R \subseteq \mathcal{C}(\mathcal{T}), \quad A_L \subseteq \mathcal{C}(\mathcal{T}^\top),$$

then

$$\dim_{\mathbb{F}_q} A_R \leq e, \quad \dim_{\mathbb{F}_q} A_L \leq e. \quad (2)$$

In particular, retaining a d -dimensional multiplier algebra on either side forces $e = d$, $\dim_{\mathbb{F}_q} \mathcal{T} = d$, and $\mathcal{T} = D \cong \mathbb{F}_{q^d}$. Every genuinely noncommutative irreducible choice has $e < d$ and therefore strictly fewer than d^2 ordered multiplier labels before any symmetry quotient is taken.

Proof. By Schur's lemma, $D = \text{End}_{\mathcal{T}}(V)$ is a finite division algebra. Wedderburn's little theorem makes it a field $\mathbb{F}_{q^e}$. The faithful D -action makes V a D -vector space, so $e \mid d$ and $r := \dim_D V = d/e$. Because the assumed irreducibility says that V is a simple $\mathcal{T}$ -module, the Jacobson density theorem identifies the image of $\mathcal{T}$ with $\text{End}_D(V)$. All spaces are finite, so density is equality [2]. Hence $\mathcal{T} = \mathcal{C}(D) \cong \text{Mat}_r(\mathbb{F}_{q^e})$ and has $\mathbb{F}_q$ -dimension $er^2 = d^2/e$.

The two alignment inclusions place A_R inside D and A_L inside $D^\Gamma = \mathcal{C}(\mathcal{T}^\Gamma)$, proving (2). Since $e \leq d$, a multiplier algebra of $\mathbb{F}_q$ -dimension d forces $e = d$ and $r = 1$. Then $\mathcal{T} = D$ is itself a degree- d field. If $\mathcal{T}$ is noncommutative, $r > 1$, so $e < d$ and the product of the left and right label dimensions is at most $e^2 < d^2$. $\square$

For $d = 4$, the alternatives are especially stark: the field extremum has a four-dimensional centralizer and sixteen ordered labels; an irreducible $\text{Mat}_2(\mathbb{F}_{q^2})$ secret algebra has only a two-dimensional centralizer and four ordered labels; and the full matrix algebra has only scalar multipliers. The irreducibility hypothesis belongs only to this capacity classification. It is not a loophole at full multiplier dimension: Proposition 3.6 below traps every pointwise aligned secret coefficient in the degree- d multiplier field, whether the generated algebra is irreducible or reducible.

Lemma 3.2 (A full-degree field is self-centralizing). *Let $B \subseteq \text{Mat}_d(\mathbb{F}_q)$ be a field of $\mathbb{F}_q$ -dimension d . Then*

$$\mathcal{C}(B) = B.$$

Proof. View $V = \mathbb{F}_q^d$ as a B -module. The embedding is faithful and $\dim_{\mathbb{F}_q} B = \dim_{\mathbb{F}_q} V = d$, so V is one-dimensional over B (and hence the B -action is automatically irreducible). An $\mathbb{F}_q$ -linear endomorphism commutes with B exactly when it is B -linear. Every B -linear endomorphism of the one-dimensional B -space V is multiplication by an element of B . $\square$

The field centralizer is also the building block for more permissive alignment in which secret mixing relabels public multipliers by a field automorphism rather than commuting with them pointwise.

Lemma 3.3 (Normalizer and uniqueness of the aligned field). *Let $d > 1$, let $B \subseteq \text{Mat}_d(\mathbb{F}_q)$ be as in Lemma 3.2, and let $\varphi : x \mapsto x^q$ denote the $\mathbb{F}_q$ -linear Frobenius map after identifying $\mathbb{F}_q^d$ with B . Then*

$$N(B) = B^\times \rtimes \langle \varphi \rangle. \tag{3}$$

Moreover, if $C \subseteq \text{Mat}_d(\mathbb{F}_q)$ is another field of $\mathbb{F}_q$ -dimension d and $C^\times \subseteq N(B)$, then $C = B$.

Proof. Conjugation by an element of $N(B)$ induces an $\mathbb{F}_q$ -automorphism of B , so it induces $x \mapsto x^{q^k}$ for some k . Composing with φ^{-k} leaves an element that centralizes B , hence lies in $B^\times$ by Lemma 3.2. This proves (3).

Choose a generator c of the cyclic group $C^\times$, so $\text{ord}(c) = q^d - 1$. Write $c = b\varphi^k$ using (3). If $k \neq 0$, put $g = \gcd(d, k)$ and $e = d/g$. Then $(b\varphi^k)^e$ is a norm lying in $\mathbb{F}_{q^g}^\times$, and therefore

$$\text{ord}(c) \mid e(q^g - 1) \leq d(q^{d/2} - 1) < q^d - 1.$$

The final strict inequality follows from $q^{d/2} + 1 > d$, which holds for $q \geq 2$ and $d > 1$. This contradicts $\text{ord}(c) = q^d - 1$. Thus $k = 0$, so $c \in B^\times$ and $C = \mathbb{F}_q[c] = B$. $\square$

A more permissive design might choose secret coefficients from a linear space that normalizes the multiplier field, so that moving a multiplier past a secret coefficient applies a Frobenius automorphism rather than fixing the multiplier pointwise. Even this freedom is rigid at the SNOVA dimensions.

Lemma 3.4 (Linear normalizer support is one semilinear coset). *Let $A \subseteq \text{Mat}_d(\mathbb{F}_q)$ be a degree- d field and let $\varphi : x \mapsto x^q$ be Frobenius in an A -coordinate model. If $0 \neq U \subseteq \text{Mat}_d(\mathbb{F}_q)$ is an $\mathbb{F}_q$ -linear subspace satisfying*

$$U \setminus \{0\} \subseteq N(A), \quad (4)$$

then, for every $0 \neq u_0 \in U$,

$$Uu_0^{-1} \subseteq A, \quad U \subseteq Au_0 = A\varphi^k \text{ for one } 0 \leq k < d, \quad \dim_{\mathbb{F}_q} U \leq d. \quad (5)$$

If $\dim U = d$, then $U = A\varphi^k$. If additionally $1 \in U$, then $U \subseteq A$, with equality when $\dim U = d$.

Now let U have dimension d . If $U \setminus \{0\}$ normalizes a right multiplier field A_R and $U^\top \setminus \{0\}$ normalizes a left multiplier field A_L , then

$$A_L = A_R^\top. \quad (6)$$

Thus full-dimensional automorphism-relabeling support still determines the same adjoint multiplier pair; it can only twist the secret support by one Frobenius-semilinear coset. No relation between q and d is required.

Proof. The standard linearized-polynomial representation gives the direct-sum decomposition

$$\text{End}_{\mathbb{F}_q}(A) = \bigoplus_{k=0}^{d-1} A\varphi^k. \quad (7)$$

Indeed, the maps $x \mapsto \sum_{k=0}^{d-1} a_k x^{q^k}$ are linearly independent: a nonzero such polynomial has ordinary degree at most $q^{d-1} < q^d$ and therefore cannot vanish on all q^d elements of A . Their total $\mathbb{F}_q$ -dimension is d^2 , so they span every $\mathbb{F}_q$ -linear endomorphism. Together with (3), this says that every nonzero normalizer element lies in exactly one summand $A\varphi^k$.

Fix $0 \neq u_0 \in U$ and put $V = Uu_0^{-1}$. Then V is an $\mathbb{F}_q$ -linear subspace, $1 \in V$, and $V \setminus \{0\} \subseteq N(A)$. Suppose that $v = a\varphi^k \in V$ with $k \neq 0$. By the direct sum (7), $1 + v$ is nonzero and has nonzero components in two distinct summands. Hence $1 + v \notin N(A)$, contradicting $1 + v \in V$. Thus every element of V lies in A , proving $Uu_0^{-1} \subseteq A$. Writing $u_0 = a_0\varphi^k$ now gives $U \subseteq A\varphi^k$ and all remaining one-sided conclusions.

For the two-sided statement, the first part and $\dim U = d$ give $Uu_0^{-1} = A_R$. Since u_0 normalizes A_R , this also gives $u_0^{-1}U = A_R$. Applied to $U^\top$, the same argument gives $A_L = U^\top u_0^{-\top}$. Therefore

$$A_L = U^\top u_0^{-\top} = (u_0^{-1}U)^\top = A_R^\top,$$

which proves (6). □

Lemma 3.5 (Large nonlinear normalizer support is rigid). *Let $A, C \subseteq \text{Mat}_d(\mathbb{F}_q)$ be degree- d field algebras. If $A \neq C$, then*

$$|N(A) \cap N(C)| \leq d^2(q^{d-1} - 1). \quad (8)$$

Consequently, let $\mathcal{S} \subseteq \text{Mat}_d(\mathbb{F}_q)$ be an arbitrary not-necessarily-linear support such that every nonzero $t \in \mathcal{S}$ normalizes a right multiplier field A_R and $t^\top$ normalizes a left multiplier field A_L . If

$$|\mathcal{S} \setminus \{0\}| > d^2(q^{d-1} - 1), \quad (9)$$

then $A_L = A_R^\top$. In particular, a support of cardinality q^d containing zero forces the adjoint pair whenever $q \geq d^2$.

Proof. Write

$$N(A) = \bigcup_{k=0}^{d-1} (A\varphi^k \setminus \{0\}), \quad N(C) = \bigcup_{j=0}^{d-1} (C\psi^j \setminus \{0\})$$

using Lemma 3.3. The displayed components are d -dimensional $\mathbb{F}_q$ -linear subspaces after adjoining zero. If $A\varphi^k = C\psi^j$ for some pair (k, j) , choose a nonzero element u_0 in that common space. Right-normalizing the common space by u_0^{-1} gives both A and C , so $A = C$. Hence, when $A \neq C$, every pairwise intersection has dimension at most $d - 1$ and contains at most $q^{d-1} - 1$ nonzero elements. A union bound over the d^2 pairs proves (8).

Put $C = A_L^\top$. If $t^\top$ normalizes A_L , transposing the conjugation identity shows that t normalizes C . Thus $\mathcal{S} \setminus \{0\} \subseteq N(A_R) \cap N(C)$. The strict inequality (9) forces $A_R = C = A_L^\top$. Finally, when $|\mathcal{S}| = q^d$ and $0 \in \mathcal{S}$, the required strict inequality follows from

$$q^d - 1 - d^2(q^{d-1} - 1) = q^{d-1}(q - d^2) + (d^2 - 1) > 0$$

for $q \geq d^2$. □

Proposition 3.6 (Pointwise full-degree alignment is rigid). *Let $A_L, A_R \subseteq \text{Mat}_d(\mathbb{F}_q)$ be degree- d field algebras and let $U \subseteq \text{Mat}_d(\mathbb{F}_q)$ be an $\mathbb{F}_q$ -linear secret-coefficient support. If*

$$t^\top a_L = a_L t^\top, \quad a_R t = t a_R \tag{10}$$

for every $t \in U$, $a_L \in A_L$, and $a_R \in A_R$, then

$$U \subseteq A_R \cap A_L^\top, \quad \dim_{\mathbb{F}_q} U \leq d. \tag{11}$$

If $\dim_{\mathbb{F}_q} U = d$, then

$$U = A_R = A_L^\top. \tag{12}$$

This conclusion needs neither irreducibility of the algebra generated by U nor closure of U under multiplication.

Proof. By Lemma 3.2, the degree- d field A_R is self-centralizing. The right commutation in (10) therefore puts every $t \in U$ in A_R . The left commutation puts $t^\top$ in $\mathcal{C}(A_L) = A_L$, hence $t \in A_L^\top$. This proves (11). If $\dim U = d$, equality of dimensions forces (12). □

The alignment condition is visible on one central summand. Let t be a secret block coefficient, let a_L and a_R be public left and right multipliers, and substitute $X = TU$. One obtains terms of the form

$$U_i^\top t^\top a_L F a_R s U_j.$$

To reuse one congruence-transformed public form while keeping the public multipliers outside it, the derivation moves a_L past $t^\top$ and a_R past s . The official construction achieves this by drawing all of these coefficients from one symmetric commutative field [3]. The general consequence is stronger.

Theorem 3.7 (Full alignment determines the multiplier pair). *Let $A_L, A_R \subseteq \text{Mat}_d(\mathbb{F}_q)$ be degree- d field algebras. The multiplier algebras are necessarily an adjoint pair under any of the following alignment hypotheses:*

- (i) a d -dimensional $\mathbb{F}_q$ -linear secret support U satisfies pointwise alignment (10);
- (ii) a d -dimensional $\mathbb{F}_q$ -linear support U has every nonzero element normalizing A_R and every nonzero element of $U^\top$ normalizing A_L ; or

(iii) an arbitrary support $\mathcal{S}$ satisfies the same two-sided normalizer condition and $|\mathcal{S} \setminus \{0\}| > d^2(q^{d-1} - 1)$.

More precisely,

$$A_L = B^\top, \quad A_R = B \quad (13)$$

for the degree- d field $B = A_R$. In case (i), $U = B = A_L^\top$. In case (ii), $U = B\varphi^k$ for one Frobenius power φ^k . Case (iii) requires neither linearity nor algebra closure of the support. No irreducibility or q -versus- d hypothesis is needed.

Proof. Case (i) is [Proposition 3.6](#), case (ii) is [Lemma 3.4](#), and case (iii) is [Lemma 3.5](#). Each gives $A_L = A_R^\top$; putting $B = A_R$ gives (13). $\square$

Thus choosing different left and right algebras does not make them independent: full alignment forces the adjoint pair $B^\top$ and B . The official same-algebra construction is the special case in which this pair coincides.

Corollary 3.8 (Same-algebra alignment forces transpose stability). *If the secret coefficients and both multiplier families use one degree- d field algebra A , then full alignment forces*

$$A^\top = A. \quad (14)$$

If pointwise public multipliers span A but one insists on $A^\top \neq A$, every alignable secret coefficient lies in the proper subspace $A \cap A^\top$.

Proof. The first claim is (13) with $A_L = A_R = B = A$. For the second, an alignable $t \in A$ has $t^\top \in \mathcal{C}(A) = A$, so $t \in A \cap A^\top$. $\square$

When (14) holds, transpose induces

$$\sigma : A \longrightarrow A, \quad \sigma(a) = a^\top.$$

Because A is commutative, this anti-automorphism is an automorphism.

Corollary 3.9 (Classification of the same-algebra adjoint). *The map σ is an $\mathbb{F}_q$ -automorphism of $A \cong \mathbb{F}_{q^d}$ satisfying $\sigma^2 = 1$. Consequently*

$$\sigma = \text{id}, \quad \text{or, when } d \text{ is even, } \sigma(x) = x^{q^{d/2}}.$$

In particular, when d is odd, full same-algebra alignment forces every element of A to be symmetric. Changing the generator or basis cannot produce any other adjoint.

Proof. For $a, b \in A$, $\sigma(ab) = b^\top a^\top = \sigma(a)\sigma(b)$, and transpose is $\mathbb{F}_q$ -linear and involutory. The $\mathbb{F}_q$ -automorphism group of $\mathbb{F}_{q^d}$ is cyclic of order d , generated by Frobenius. It has no nontrivial element of order two when d is odd and has the unique such element $x \mapsto x^{q^{d/2}}$ when d is even. $\square$

4 The Hermitian quotient forced by alignment

Assume now that q is odd. Under full alignment, write $A_L = B^\top$ and $A_R = B$ as in (13). Define an involution on the ordered left/right label space by

$$\tau(\xi \otimes \eta) = \eta^\top \otimes \xi^\top, \quad \xi \in A_L, \eta \in A_R. \quad (15)$$

The Hermitian and anti-Hermitian label quotients are

$$\text{Her}^2(A_L, A_R) = (A_L \otimes_{\mathbb{F}_q} A_R) / \text{span}\{z - \tau z : z \in A_L \otimes A_R\}, \quad (16)$$

$$\text{AHer}^2(A_L, A_R) = (A_L \otimes_{\mathbb{F}_q} A_R) / \text{span}\{z + \tau z : z \in A_L \otimes A_R\}. \quad (17)$$

For a base matrix P , put

$$\Phi_P(\xi \otimes \eta)(u) = u^\top \Sigma_\xi P \Sigma_\eta u.$$

Theorem 4.1 (Alignment–Hermitian factorization). *For every base matrix P ,*

$$\Phi_P \circ \tau = \Phi_{P^\top}. \quad (18)$$

Consequently:

- (i) if $P^\top = P$, then Φ_P factors through $\text{Her}^2(A_L, A_R)$;
- (ii) if $P^\top = -P$, then Φ_P factors through $\text{AHer}^2(A_L, A_R)$;
- (iii) the quotient dimensions are

$$\dim_{\mathbb{F}_q} \text{Her}^2(A_L, A_R) = \frac{d(d+1)}{2}, \quad \dim_{\mathbb{F}_q} \text{AHer}^2(A_L, A_R) = \frac{d(d-1)}{2}. \quad (19)$$

Thus every fully aligned symmetric redesign has the same universal $d(d+1)/2$ cap, even if it uses distinct left and right algebras; additional dependencies can only lower the rank.

Proof. The block embedding is transpose-compatible: $\Sigma_{a^\top}^\top = \Sigma_a$. Therefore, for a pure tensor, transposing the scalar gives

$$\begin{aligned} \Phi_P(\tau(\xi \otimes \eta))(u) &= u^\top \Sigma_{\eta^\top} P \Sigma_{\xi^\top} u \\ &= u^\top \Sigma_\xi P^\top \Sigma_\eta u = \Phi_{P^\top}(\xi \otimes \eta)(u), \end{aligned}$$

which proves (18). The two factorization statements follow.

Choose a basis $b_1, \dots, b_d$ of B and the corresponding basis $b_1^\top, \dots, b_d^\top$ of $B^\top$. In the basis $b_i^\top \otimes b_j$, the involution τ is ordinary swap:

$$b_i^\top \otimes b_j \mapsto b_j^\top \otimes b_i.$$

Because q is odd, the tensor product is the direct sum of the $+1$ and -1 eigenspaces. They have dimensions $(d^2 + d)/2$ and $(d^2 - d)/2$, which are the dimensions of the two quotients. $\square$

Corollary 4.2 (Noncommutative alignment only lowers symmetric capacity). *Let $\mathcal{T}$ be any irreducible secret algebra from Theorem 3.1, with centralizer $D \cong \mathbb{F}_{q^e}$. Under pointwise alignment using the full centralizer pair $D^\top, D$, every symmetric base form has ordered-label rank at most*

$$\frac{e(e+1)}{2}. \quad (20)$$

The raw label space has dimension only e^2 . Hence the field extremum $e = d$ simultaneously maximizes the aligned multiplier capacity and maximizes the symmetric quotient; every genuinely noncommutative irreducible secret algebra has a strictly smaller cap.

Proof. Choose a basis $b_1, \dots, b_e$ of D and the transpose basis of $D^\top$. On the e^2 labels $b_i^\top \otimes b_j$, scalar transposition again acts by ordinary swap. A symmetric form therefore identifies (i, j) with (j, i) and leaves at most $e(e+1)/2$ directions. The strict comparison follows from $e < d$ in the noncommutative case of Theorem 3.1. $\square$

Table 1: The alignment-capacity alternatives for the public block sizes. The final column is the symmetric-form cap when the full centralizer pair is used. Noncommutativity increases secret-algebra dimension only by shrinking the public multiplier labels.

block d	irreducible secret algebra $\mathcal{T}$	e	$\dim \mathcal{T}$	raw labels	symmetric cap
2	$\mathbb{F}_{q^2}$	2	2	4	3
2	$\text{Mat}_2(\mathbb{F}_q)$	1	4	1	1
4	$\mathbb{F}_{q^4}$	4	4	16	10
4	$\text{Mat}_2(\mathbb{F}_{q^2})$	2	8	4	3
4	$\text{Mat}_4(\mathbb{F}_q)$	1	16	1	1

In the official same-algebra case $A_L = A_R = A$, the adjoint σ from [Corollary 3.9](#) rewrites (15) as

$$\tau(\xi \otimes \eta) = \sigma(\eta) \otimes \sigma(\xi).$$

When $\sigma = \text{id}$, the two quotients are the ordinary symmetric and alternating squares. The theorem therefore rules out three superficial repairs at once. A nonsymmetric generator does not help if same-algebra alignment is retained; distinct left and right algebras are forced to be an adjoint pair and give the same quotient; and increasing q does not change the dimensions.

5 How much nonsymmetry is required to escape

The preceding theorem does not claim that every arbitrary nonsymmetric base family is already broken. It instead quantifies how radical such a repair must be. Let W be a w -dimensional space of base matrices and define

$$W_+ = \text{span}\{P + P^\top : P \in W\}, \quad W_- = \text{span}\{P - P^\top : P \in W\}, \quad s_\pm = \dim W_\pm.$$

Let $\mathcal{I}_+$ and $\mathcal{I}_-$ be the polynomial image spaces generated by W_+ and W_- , respectively, over all aligned ordered labels.

Theorem 5.1 (Hermitian-channel rank bound). *For the complete feature map generated by W and the aligned ordered label space $A_L \otimes A_R$,*

$$\text{rank } \Phi_W \leq s_+ \frac{d(d+1)}{2} + s_- \frac{d(d-1)}{2}. \quad (21)$$

In particular:

(a) *if W is transpose-stable, then $s_+ + s_- = w$ and*

$$\text{rank } \Phi_W \leq w \frac{d(d+1)}{2} < wd^2 \quad (d > 1); \quad (22)$$

(b) *attaining the raw ordered-label rank wd^2 requires*

$$s_+ = s_- = w. \quad (23)$$

Hence both symmetrization and skew-symmetrization must be injective on W . Equality also forces

$$\dim \mathcal{I}_+ = w \frac{d(d+1)}{2}, \quad \dim \mathcal{I}_- = w \frac{d(d-1)}{2}, \quad \mathcal{I}_+ \cap \mathcal{I}_- = \{0\}, \quad (24)$$

and necessarily

$$W \cap W^\top = \{0\}. \quad (25)$$

Thus a full-rank nonsymmetric repair cannot be a small perturbation of a symmetric or transpose-stable family: it must be maximally non-self-adjoint and must expose full, mutually disjoint Hermitian and anti-Hermitian channels. Moreover, $W \cap W^\top = 0$ forces

$$\dim(W + W^\top) = 2w. \quad (26)$$

Hence the escape doubles the dimension of the transpose-stable closure: its canonical decomposition contains w independent Hermitian and w independent anti-Hermitian base directions.

Proof. Because 2 is invertible, $P = (P + P^\top)/2 + (P - P^\top)/2$. Apply Theorem 4.1 to the two summands and then sum their image-dimension bounds. This proves (21).

If W is transpose-stable, it is the direct sum of its symmetric and skew-symmetric eigenspaces, so $s_+ + s_- = w$. Since $d(d+1)/2 \geq d(d-1)/2$, (22) follows. For arbitrary W , both s_+ and s_- are at most w . Equality in the upper bound $wd^2 = w(d(d+1)/2 + d(d-1)/2)$ therefore requires (23). Moreover, $\text{im } \Phi_W \subseteq \mathcal{I}_+ + \mathcal{I}_-$, while

$$\dim \mathcal{I}_+ \leq s_+ \frac{d(d+1)}{2}, \quad \dim \mathcal{I}_- \leq s_- \frac{d(d-1)}{2}.$$

Equality $\text{rank } \Phi_W = wd^2$ forces equality in both image bounds and in the dimension formula for $\mathcal{I}_+ + \mathcal{I}_-$, proving (24).

Suppose now that $0 \neq P \in W \cap W^\top$. Write $P = Q^\top$ with $Q \in W$. If $P + Q \neq 0$, then $P + Q$ is a nonzero symmetric element of W , contradicting injectivity of skew-symmetrization. If $P + Q = 0$, then P is a nonzero skew-symmetric element of W , contradicting injectivity of symmetrization. Hence (25) is necessary. Finally,

$$\dim(W + W^\top) = \dim W + \dim W^\top - \dim(W \cap W^\top) = 2w,$$

which proves (26). □

Corollary 5.2 (Alignment–Hermitian no-go theorem). *Consider an odd-characteristic SNOVA-style redesign that retains:*

- (i) degree- d aligned left/right multiplier fields and a secret block support that is either a full d -dimensional linear space or has more than $d^2(q^{d-1} - 1)$ nonzero coefficients;
- (ii) key-randomness alignment through the standard congruence-reuse identity, either by pointwise commutation or by multiplier-field automorphism relabeling; and
- (iii) a security count of wd^2 ordered power-pair features from w base forms.

Then no reducible or noncommutative choice of secret support, generator change, basis change, field-size increase, distinct aligned multiplier pair, or transpose-stable base-family modification can validate that count. A repair must choose at least one of the following departures:

1. retain symmetric or transpose-stable forms and accept the Hermitian cap in (22);
2. replace them by a maximally non-self-adjoint family satisfying (23)–(25), including the full disjoint-channel condition (24), thereby restoring a full anti-Hermitian (exterior when $\sigma = 1$) channel; or
3. abandon full alignment or change the public-key derivation so that the congruence-transformed forms can no longer be reused in the same way.

The companion cryptanalysis gives explicit forgery systems for all nine Version 2.3 rows [1]. The theorem does not assert that every construction in the second or third branch is already broken; it proves that either branch is a fundamental architectural change rather than a local repair of the present philosophy.

Proof. Full alignment determines the adjoint multiplier pair by [Theorem 3.7](#). The symmetric and transpose-stable branches are bounded by [Theorems 4.1](#) and [5.1](#). Achieving the full ordered rank forces the second branch by [Theorem 5.1](#). Refusing the adjoint-pair structure forces abandonment of full alignment by [Theorem 3.7](#). $\square$

The anti-Hermitian channel is not an invented bookkeeping device. When $\sigma = 1$ it is exactly $\wedge^2 A$, the exterior channel that appears in wedge and exterior-algebra cryptanalysis; the complementary Hermitian channel is the symmetric-algebra side studied in odd-characteristic UOV analysis [\[4, 5, 6\]](#). Those prior results do not by themselves prove every maximally non-self-adjoint redesign insecure. They do explain why restoring the missing channel is not the same as restoring generic random-MQ behavior.

6 Application to SNOVA Version 2.3

For one base form, the ordered label space is $A \otimes_{\mathbb{F}_q} A$. The following corollary specializes the Hermitian quotient to Version 2.3.

Corollary 6.1 (Symmetric-square factorization). *Let $A = \mathbb{F}_q[S]$ have dimension d , and assume $S^\top = S$ and $P_i^\top = P_i$. For each i , the map*

$$A \otimes_{\mathbb{F}_q} A \longrightarrow \mathbb{F}_q[u]_2^{\text{hom}}, \quad \xi \otimes \eta \longmapsto u^\top \Sigma_\xi P_i \Sigma_\eta u$$

is invariant under $\xi \otimes \eta \mapsto \eta \otimes \xi$. It factors through

$$\text{Sym}_{\mathbb{F}_q}^2(A) = (A \otimes_{\mathbb{F}_q} A) / \text{span}\{\xi \otimes \eta - \eta \otimes \xi\}.$$

Consequently the direct sum over m_1 base forms has rank at most $m_1 \binom{d+1}{2}$. After any public linear map to M outputs, its rank is at most

$$\min \left\{ M, m_1 \binom{d+1}{2} \right\}. \quad (27)$$

If the minimal polynomial of S has degree ℓ , then $d = \ell$ and the ordered feature vector is obtained by duplicating the off-diagonal coordinates of an unordered feature vector.

Proof. Here $\sigma = \text{id}$, so the claim follows directly from [Theorem 4.1](#). Concretely, every element of A is symmetric. Since the displayed value is a scalar,

$$u^\top \Sigma_\xi P_i \Sigma_\eta u = (u^\top \Sigma_\xi P_i \Sigma_\eta u)^\top = u^\top \Sigma_\eta P_i \Sigma_\xi u.$$

Thus every alternating label $\xi \otimes \eta - \eta \otimes \xi$ lies in the kernel. The symmetric square has dimension $\binom{d+1}{2}$, and a later linear map cannot enlarge its image. When powers of S form a basis, the coordinate statement is exactly the duplication of each off-diagonal unordered pair into its two ordered positions. $\square$

The corollary gives an upper bound: extra public dependencies can only reduce the quadratic image further. Every concrete attack checks the exact public rank before solving. The outer map cannot repair the loss because it sees only the already-equal polynomial values. The factorization itself uses only transposition symmetry. Odd characteristic is needed later for the familiar decomposition into symmetric and alternating tensors and for the polar-form analysis. The concrete

symmetric-square corollary and the odd-characteristic forgery analysis do not cover the submitted nonsymmetrized $q = 16$ map. The alignment theorem still diagnoses its field-algebra architecture, but the characteristic-two label decomposition and its wedge attack are different.

The published construction uses a single algebra $A = \mathbb{F}_q[S] \subseteq \text{Mat}_d(\mathbb{F}_q)$ with $S^\top = S$ and symmetric base matrices. Full same-algebra alignment therefore has trivial adjoint on A , and the Hermitian quotient is the ordinary symmetric square. One base form contributes at most

$$\binom{d+1}{2}$$

independent quadratic features, rather than the d^2 ordered labels used in the unquotiented count. Thus

$$d = 4 : \quad 16 \longrightarrow 10, \quad d = 2 : \quad 4 \longrightarrow 3.$$

The loss occurs before any outer linear mixing and cannot be repaired by that mixing.

The companion cryptanalysis [1] turns this identity into complete square forgery systems for every published $q = 19$ Version 2.3 parameter set. That paper contains the exact affine reduction, rejection handling, root-spectrum analysis, symbolic-homotopy recovery, streamed-XL alternatives, and concrete numerical ledgers. None of those attack details are needed for the architectural theorem here; conversely, the concrete break does not depend on the broader no-local-repair theorem.

The separation is useful. The concrete paper answers whether the published scheme survives. This paper answers which parts of the surrounding design philosophy can be retained in a repair. Keeping full alignment and a transpose-stable base family forces the Hermitian cap. Recovering the raw ordered-label count requires a maximally non-self-adjoint family and a full anti-Hermitian channel, or else a different key-derivation mechanism.

7 Relation to prior work and limitations

The anti-Hermitian channel is the exterior square when the field adjoint is trivial. Exterior and wedge channels already play a central role in cryptanalysis of SNOVA and related UOV-style systems [6, 7]. Complementary symmetric channels arise in odd-characteristic UOV analyses [4, 5]. Our contribution is not a new solver for those channels. It is a rigidity theorem explaining why full-capacity alignment canonically produces the adjoint pair and why transpose-stable public forms cannot support the raw ordered-label count.

Several boundaries are important. First, a maximally non-self-adjoint family may evade the Hermitian cap; our theorem gives necessary conditions for full capacity, not a cryptanalysis of every family satisfying them. Second, a design may abandon full alignment, accept a larger public key, or use a key-derivation identity outside the congruence-reuse model. Third, in characteristic two the symmetric/skew eigenspace decomposition changes and must be analyzed through the appropriate exterior-channel formalism. These are genuine departures, not contradictions of the theorem.

8 Conclusion

Full-capacity key-randomness alignment is far more rigid than a choice of basis or generator. It forces degree- d multiplier fields into an adjoint pair, survives Frobenius-semilinear relabeling and large nonlinear supports, and is uniquely maximized by the field case among irreducible secret algebras. In odd characteristic, the adjoint pair splits ordered quadratic labels into Hermitian and

anti-Hermitian sectors. Symmetric or transpose-stable public forms cannot realize the raw d^2 feature count.

Accordingly, a SNOVA-style repair cannot simultaneously preserve full alignment, congruence reuse, transpose-stable forms, and the ordered-label security rationale. It must accept the smaller quotient, restore a full and disjoint anti-Hermitian channel through a maximally non-self-adjoint base family, or replace the alignment mechanism. The companion paper shows that the published Version 2.3 design, which takes the first branch, admits the explicit all-nine forgery systems analyzed there.

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